

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester I)

Scala For DS-8

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Class :FYIT	Batch : 2
Date of Assignment :21-07-2026	Date/Time of Submission :21/07/26

Practical 8

Aim: Slice a breeze matrix to extract a sub-matrix and calculate its row and column Sums.

Code:

```
import breeze.linalg._
object Pract8 extends App {
  val m =
DenseMatrix((1.0,2.0,3.0,4.0), (5.0,6.0,7.0,8.0), (9.0,10.0,11.0,12.0), (13.0,
14.0,15.0,16.0))

  val s = m(1 to 2, 1 to 2)
  val r = sum(s(*, :))
  val c = sum(s(:, *))

  println(s"Original Matrix : \n$m")
  println(s"\nSub-Matrix : \n$s")
  println(s"\nRow Sums: $r")
  println(s"Column Sums : $c")

  println("Himanshu S037")
}
```

Output:

Original Matrix :

1.0	2.0	3.0	4.0
5.0	6.0	7.0	8.0
9.0	10.0	11.0	12.0
13.0	14.0	15.0	16.0

Sub-Matrix :

6.0	7.0
10.0	11.0

Row Sums: DenseVector(13.0, 21.0)

Column Sums : Transpose(DenseVector(16.0, 18.0))

Himanshu S037